

EXPERIMENT 9

Classification of Physiological Signals (ECG) using Machine Learning Models

1. Problem Definition

The objective of this experiment is to classify ECG (electrocardiogram) signals into:

- **0 → Normal**
- **1 → Abnormal**

This is a **binary classification problem**, where the goal is to detect abnormal heart conditions using machine learning models.

2. Data Understanding

Two datasets were used:

1. MIT-BIH Dataset

- Originally multi-class : converted to binary
- Normal: class 0
- Abnormal: classes 1–4

2. PTBDB Dataset

- Already binary:
 - 0 : Normal
 - 1 : Abnormal

Dataset Characteristics

- Features: **187 (ECG signal values)**
- Samples: **~21,892 (MIT-BIH test set)**

- MIT-BIH is **imbalanced**
- Normal signals dominate abnormal ones
- Accuracy can be misleading
- Need to analyze recall and confusion matrix

3. Data Preprocessing

Steps applied:

1. Converted MIT-BIH to binary classification
2. Combined PTBDB normal + abnormal datasets
3. Feature-target split
4. Feature scaling using **StandardScaler**

Important:

- Scaling is necessary for **KNN and SVM**
- Without scaling : poor distance calculations

4. Models Used

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)

5. Evaluation and Prediction

- Accuracy
- Precision
- Recall
- F1-score

- Confusion Matrix

Important:

- In medical problems : **Recall is critical**
- Missing abnormal cases = real-world failure

MIT-BIH Dataset Results

Accuracy Comparison

Model	Accuracy
Logistic Regression	0.9046
KNN	0.9766
Decision Tree	0.9596
Random Forest	0.9775 (Best)
SVM	0.9701

Confusion Matrix Analysis

Logistic Regression

[[17727 391]
[1697 2077]]

Problem:

- **1697 abnormal cases missed (FN)**
- Very dangerous in medical context

KNN

[[17999 119]
[393 3381]]

Much better:

- High TP (detects abnormal cases well)

- Lower false negatives

Decision Tree

[[17670 448]
[436 3338]]

Moderate:

- More errors than KNN

Random Forest (BEST)

[[18090 28]
[436 3338]]

Insight:

- Almost zero false positives
- Strong detection of abnormal signals
- Best overall balance

PTBDB Dataset Results

Accuracy Comparison

Model	Accuracy
Logistic Regression	0.8337
KNN	0.9296
Decision Tree	0.9255
Random Forest	0.9722 (Best)
SVM	0.9189

6. Model Comparison Analysis

Key Observations

1. Logistic Regression Performs Poorly

- Lowest accuracy in both datasets
- Misses many abnormal cases

Reason:

- ECG data is highly **non-linear**
- Linear model fails

2. Random Forest is Consistently Best

- Highest accuracy in both datasets
- Strong performance across all classes

Reason:

- Handles complex patterns
- Reduces overfitting

3. KNN Performs Surprisingly Well

- Second-best performance
- Works well due to pattern similarity in ECG signals

4. Decision Tree is Unstable

- Performs worse than ensemble methods
- Prone to overfitting

5. SVM Performs Well but Not Best

- Good accuracy
- Slightly worse than Random Forest

Best Model Selection

Random Forest is the best model, because:

- Highest accuracy
- Strong recall
- Robust across datasets